

Gaurab Pokharel

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Research interests

Algorithmic fairness, game theory, and machine learning for the allocation of scarce societal resources; interpretable models of human discretion in public-sector decision-making; evaluation and reliability of large language models for high-stakes judgments; learning dynamics in two-sided matching markets.

Education

PhD (Computer Science) <i>Virginia Tech</i>	2024 – present
M.Sc. (Computer Science) <i>George Mason University</i>	2021 – 2023 CGPA: 3.96/4.0
B.A. (Computer Science and Mathematics) <i>Oberlin College</i>	2017 – 2021 CGPA: 3.82/4.0

Experience

Associate AI Scientist, Intern May 2026 – Aug 2026
Pearson

- Designed a two-arm study to test procedural equivalence in constructing rater-calibration “training packs” for a state reading assessment: an observational contrast of historical human-built versus current model-assisted packs, and a prospective controlled comparison across human-only, model-assisted, and consensus-gold processes; framed the question as a non-inferiority estimand and specified deconfounding strategies to unalias the manual-vs-model contrast from year.
- Developed a response-agnostic Wasserstein distance over ordinal score-band labels to compare pack profiles without requiring shared responses, and built the surrounding inference framework (TOST equivalence testing with a margin scaled to observed human pack spread, estimated in a mixed model with item and rater random effects).
- Applied post-hoc interpretability methods (integrated gradients, counterfactual perturbation, probing classifiers, collinearity-stable relative-importance analysis) to a proprietary transformer scorer with no exposed internals, characterizing the linguistic cues associated with score assignments.
- Engineered a reproducible analysis pipeline (Python 3.12) with a synthetic-data generator, a single configuration flag to switch between simulated and production data, pinned dependency lockfiles, and drift-blocking pre-commit hooks; validated the full pipeline on simulated data ahead of data-collection approval.
- Authored a technical methods note and a pre-registration of the primary estimand and equivalence margin, and produced operational briefing materials for experienced scorers and an independent gold panel; presented the study design to research and operations stakeholders.

Graduate Research Assistant Aug 2021 – present
Advisor: Sanmay Das

- Built and validated models of real-world human decision-making using large, noisy administrative datasets; emphasized replicable pipelines, clear error analysis, and decision-relevant metrics.
- Developed interpretable ML baselines (short decision trees) to approximate “default” judgment policies and quantified systematic deviations; used this framework to study consistency and outcome correlates of overrides.
- Designed evaluation protocols for AI judgment systems (including repeated-run robustness checks, aggregation from pairwise preferences to global rankings, and structured failure-mode analysis).
- Implemented simulation and sequential decision-making methods (bandits / learning dynamics) relevant to adaptive sampling strategies and resource allocation under uncertainty.
- Produced well-documented Python tooling for data preparation, feature generation, model training, and reporting to support iterative experimentation with strict temporal separation.

- Supported instruction for graduate ML (office hours, grading, exam logistics); strengthened communication of technical ideas and careful evaluation practice.

Publications

Peer-reviewed conference and journal publications

- [1] **G. Pokharel**, D. Sen, S. Das, and J. Ziani. “Fixed Points and Stochastic Meritocracies: A Long-Term Perspective”. In: *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency*. FAccT ’26. Montreal, Canada: Association for Computing Machinery, 2026, pp. 280–320. doi: 10.1145/3805689.3806540.
- [2] **G. Pokharel**, S. Farabi, P. J. Fowler, and S. Das. “Street-Level AI: Are Large Language Models Ready for Real-World Judgments?”. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 8.3 (Oct. 2025), pp. 2043–2054. doi: 10.1609/aies.v8i3.36694. URL: <https://ojs.aaai.org/index.php/AIES/article/view/36694>.
- [3] **G. Pokharel**, S. Das, and P. Fowler. “Discretionary Trees: Understanding Street-Level Bureaucracy via Machine Learning”. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 38.20 (Mar. 2024), pp. 22303–22312. doi: 10.1609/aaai.v38i20.30236. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/30236>.

Preprints and working papers

- [4] **G. Pokharel**, S. Das, and P. J. Fowler. “Budgeting Discretion: Theory and Evidence on Street-Level Decision-Making”. In: (2026). arXiv:2602.10039. doi: 10.48550/ARXIV.2602.10039. URL: <https://arxiv.org/abs/2602.10039>.
- [5] **G. Pokharel**, S. Farabi, P. J. Fowler, and S. Das. “Can LLMs Rank? A Tale of Triads and Triage”. In: (June 2026). arXiv:2606.30412. doi: 10.48550/arXiv.2606.30412. URL: <http://arxiv.org/abs/2606.30412>.
- [6] **G. Pokharel** and S. Das. “Converging to Stability in Two-Sided Bandits: The Case of Unknown Preferences on Both Sides of a Matching Market”. In: (Feb. 2023). arXiv:2302.06176. doi: 10.48550/arXiv.2302.06176. URL: <http://arxiv.org/abs/2302.06176>.

Extended abstracts and doctoral consortium

- [7] **G. Pokharel**. “Beyond Automation: Understanding Fairness, Ethics, and Human Discretion in AI-driven Societal Decisions”. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* 8.3 (Oct. 2025), pp. 2918–2920. doi: 10.1609/aies.v8i3.36793. URL: <https://ojs.aaai.org/index.php/AIES/article/view/36793>.

Talks and presentations

Fixed Points and Stochastic Meritocracies: A Long-Term Perspective <i>Paper presentation, ACM FAccT 2026, Montreal, Canada</i>	June 2026
Converging to Stability in Two-Sided Bandits <i>Paper presentation, Multi-Agent AI in the Real World Workshop, AAAI 2025, Philadelphia, USA</i>	March 2025
Using Machine Learning to Better Understand the Efficiency and Fairness of Allocation Systems <i>Invited Talk, DIMACS workshop on Algorithm and Mechanism Design for Achieving the UN Sustainable Development Goals, New Jersey, USA</i>	October 2023

Awards and recognitions

Best Paper , Multi-Agent AI in the Real World Workshop at AAAI	2025
Distinguished Academic Achievement Award , George Mason University	2025
The R.J. Thomas ’52 Award for Outstanding Computer Science Students , Oberlin College	2021
Oberlin College Research Fellow	2019 – 2021

Service

Reviewer AAAI ('25, '26, '27), AIES ('25, '26), EAAMO ('25, '26), FAccT ('26), EC ('26), NeurIPS ('26)

Organizing Local Arrangements Chair, CI+HCOMP '26
Workflow Chair, AIES '23
Local Chair, EAAMO '22

Technical skills

Programming Python (Expert), pandas, NumPy, scikit-learn, PyTorch, SQL, Git, Linux/Bash, $\text{\LaTeX}$

AI Evaluation Prompt Engineering, Rank Centrality, Bradley-Terry Models, ROC/AUC, Spearman Correlation, Local Inference

Modeling XGBoost, GLMs, Causal Inference (BART), Markov Decision Processes, Multi-Armed Bandits